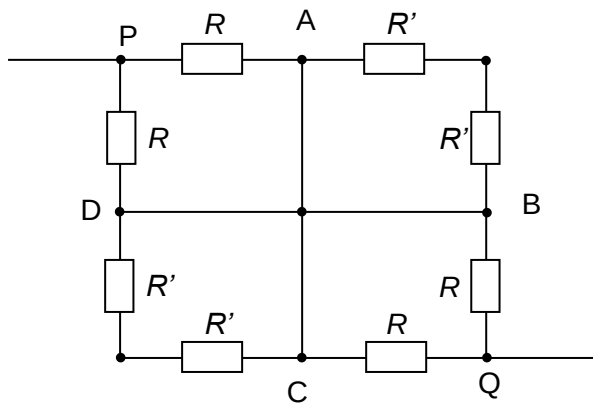


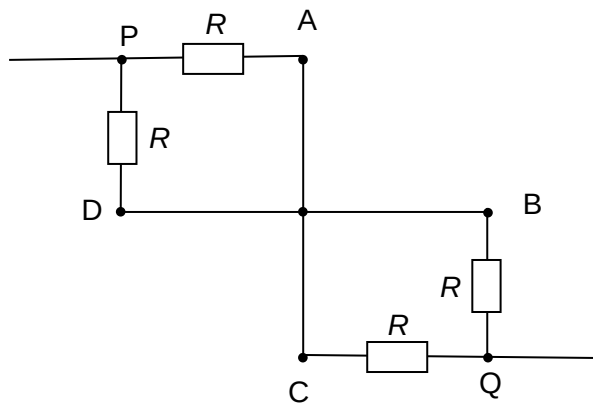
23 C



Points A, B, C and D have the same potentials.

Thus, no current flows through all 4  $R'$  resistors.

The above circuit can be re-drawn as shown below.



The effective resistance between P and Q is the twice of the net resistance of two resistors (across points P & D and P & A) in parallel. The effective resistance between P and Q =  $R$ .

**21 A** Apply RHGR to determine the direction of B-field at O due to the current flowing in a straight wire at P, O, R.